

MFG-02: Carrington Storm Motors — Supply Chain Development

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1.0 EXECUTIVE SUMMARY

Carrington Storm Motors' supply chain is built on five critical material families: MXene precursors (MAX phase, etchants), basalt fiber, carbon nanotubes/polymer resins, ceramic powders, and geopolymers precursors. Each material family has distinct supply dynamics, risks, and sourcing strategies.

This document maps the supply chain for each critical material, outlines the supplier qualification process, details the dual/multi-sourcing strategy, and addresses supply chain risk management — geopolitical, natural disaster, and supplier failure risks.

As El Segundo frames it: “The most elegant technology in the world cannot protect the grid if the supply chain cannot deliver the materials to manufacture it. Supply chain IS mission capability.”

2.0 CRITICAL MATERIALS SOURCING

2.1 MXene Precursors

Ti₃AlC₂ MAX Phase: - Current supply: Limited — few commercial producers (primarily research-grade suppliers: Carbon-Ukraine, Y-Carbon, Forsman Scientific) - Volume requirement at scale: 100-1,000 tons/year (Phase 3) - Supply risk: HIGH — nascent supply chain; limited producers - Mitigation: (1) Develop in-house MAX phase synthesis capability (TiC + Ti + Al powders, high-temperature sintering); (2) Qualify multiple commercial suppliers; (3) Fund supplier capacity expansion through long-term offtake agreements

Lithium Fluoride (LiF): - Supply: Well-established (lithium chemical industry — battery supply chain) - Volume requirement: 10-100 tons/year - Risk: MODERATE — lithium market is competitive but adequate; geopolitical concentration in Chile, Argentina, Australia, China - Mitigation: Multiple suppliers across geographies; lithium market diversification is ongoing

Hydrochloric Acid (HCl): - Supply: Commodity chemical — globally abundant - Risk: LOW — multiple producers in every industrial economy - Mitigation: Regional sourcing from nearest qualified chemical supplier

2.2 Basalt Fiber

Continuous Basalt Roving: - Supply: Established industry (basalt fiber producers: Technobasalt-Invest (Ukraine), Basalt Fiber & Composite Materials (Russia), Mafic (Ireland), Shanxi Basalt Fiber (China), Basaltex (Belgium), Neuvokas (US)) - Volume requirement: 500-5,000 tons/year at scale - Risk: MODERATE-HIGH — geopolitical concentration (Russia/Ukraine war disrupts major producers); US domestic production limited - Mitigation: (1) Qualify US/EU producers (Mafic-Ireland, Neuvokas-US, Basaltex-Belgium); (2) Dual-source across geographies; (3) Evaluate forward integration (CSM-owned basalt fiber production if supply constraints persist)

2.3 Carbon Nanotubes

Multi-Walled CNTs: - Supply: Growing industry (>50 producers globally: Nanocyl (Belgium), Arkema (France), Showa Denko (Japan), Cnano Technology (China), OCSiAl (Luxembourg/Russia), Cabot (US), LG Chem (Korea)) - Volume requirement: 10-100 tons/year at scale - Risk: MODERATE — adequate supply; quality consistency varies by producer - Mitigation: Qualify 3+ producers across US/EU/Asia; lock in specifications through quality agreements; maintain in-house CVD capability for strategic independence

Polymer Resins (Epoxy, PP, Silicone): - Supply: Commodity chemical industry — abundant - Risk: LOW — multiple global producers (Huntsman, Olin, Hexion, Dow, BASF) - Mitigation: Standard supplier qualification; safety stock; competitive bidding

2.4 Ceramic Powders

Barium Carbonate (BaCO₃): - Supply: Mineral industry — China dominates (~50% of global production) - Risk: MODERATE — geopolitical concentration - Mitigation: Multi-source from alternative producers (India, Germany, US); stockpile inventory; evaluate alternative dielectric formulations with less concentrated precursors

Titanium Dioxide (TiO₂): - Supply: Commodity pigment — abundant globally - Risk: LOW - Mitigation: Multiple suppliers

Strontium Carbonate (SrCO₃): - Supply: Specialty mineral — adequate - Risk: MODERATE - Mitigation: Multi-source

2.5 Geopolymer Precursors

Metakaolin: - Supply: Kaolin clay calcination — moderate industry - Risk: LOW-MODERATE - Mitigation: Regional sourcing; qualify multiple metakaolin producers

Fly Ash (Class F): - Supply: Coal power plant byproduct — abundant but declining as coal plants retire - Risk: LONG-TERM — fly ash availability will decrease with coal phaseout - Mitigation: Develop metakaolin-based formulations as fly ash replacement; slag-based alternatives

Ground Granulated Blast Furnace Slag (GGBFS): - Supply: Steel industry byproduct — abundant; quality varies by source - Risk: LOW-MODERATE - Mitigation: Qualify regional slag sources; develop quality adjustment protocols for slag variability

3.0 SUPPLIER QUALIFICATION PROCESS

3.1 Pre-Qualification

Step	Activity	Timeline
1. Identify	Market research to identify potential suppliers for each material	Ongoing
2. Questionnaire	Supplier self-assessment: financial stability, quality certifications (ISO 9001, IATF 16949), production capacity, environmental/social compliance	2-4 weeks
3. Sample Request	CSM requests material samples with Certificate of Analysis (CoA)	2-4 weeks
4. Testing	CSM tests samples against internal specifications	2-4 weeks
5. Audit	On-site audit of supplier facility (quality systems, production capability, EHS, labor practices)	1-3 days + report
6. Qualification	Approved Supplier List (ASL) entry; initial purchase order	1-2 weeks

3.2 Ongoing Monitoring

Metric	Frequency	Action Trigger
On-Time Delivery (OTD)	Monthly	<95% OTD triggers corrective action request
Quality (incoming inspection pass rate)	Per shipment	<98% pass rate triggers supplier corrective action
Cost competitiveness	Annual benchmarking	>10% above market triggers renegotiation or re-sourcing

Metric	Frequency	Action Trigger
Financial health	Annual (D&B, financial statements)	Deterioration triggers risk mitigation (safety stock, alternative qualification)
EHS / Social compliance	Annual audit or self-assessment	Non-compliance triggers corrective action or disqualification

4.0 DUAL/MULTI-SOURCING STRATEGY

4.1 Sourcing Strategy by Material Criticality

Criticality	Materials	Sourcing Strategy
CRITICAL (single-source would halt production)	MAX phase, LiF, specific CNT grades	Minimum 2 qualified suppliers, preferably 3, from different geographic regions
HIGH (disruption would delay production)	Basalt fiber, BaCO ₃ , specific polymer grades	2 qualified suppliers minimum
MODERATE (disruption manageable with inventory)	Metakaolin, fly ash/slag, TiO ₂ , general chemicals	2+ suppliers; regional sourcing
LOW (commodity, abundant alternatives)	HCl, general solvents, packaging, fasteners	Competitive sourcing; minimum 1 supplier

4.2 Geographic Diversification

For critical materials, CSM sources from suppliers in at least two different geographic regions to mitigate: - Regional natural disasters (earthquake, hurricane, flood) - Geopolitical disruption (trade war, sanctions, conflict) - Pandemic/health crisis affecting specific region - Logistics disruption (port closure, canal blockage, rail strike)

5.0 INVENTORY STRATEGY

5.1 Safety Stock Policy

Material Criticality	Safety Stock Target	Rationale
CRITICAL	3-6 months of demand	Long lead times; single-source risk during qualification
HIGH	2-3 months	Supplier diversification reduces but does not eliminate risk
MODERATE	1-2 months	Manageable lead times; multiple suppliers
LOW	2-4 weeks	Short lead times; abundant supply

5.2 Inventory Carrying Cost

CSM's inventory carrying cost is estimated at 20-30% of inventory value annually (capital cost + warehousing + insurance + obsolescence). The tradeoff between carrying cost and stockout risk must be optimized.

6.0 LOGISTICS AND WAREHOUSING

6.1 Warehousing Strategy

Phase	Facility Type	Locations
Phase 1	On-site warehouse at pilot facility (5,000-15,000 sq ft)	Single location
Phase 2	Central distribution center (50,000-100,000 sq ft) + on-site at manufacturing plant	Midwest or Southeast US
Phase 3	Regional distribution centers (3-5 globally) + on-site at each plant	US East, US West, EU, Asia-Pacific

6.2 Transportation Modes

Material	Mode	Rationale
MAX phase, BaCO ₃ , TiO ₂ , metakaolin, fly ash (bulk solids)	Rail or truck (bulk)	Lowest cost per ton for bulk solids
Basalt fiber roving (spools)	Truck or container ship	Medium density; damage-sensitive packaging
CNTs (low-density powder)	Air or expedited truck	High value per kg; low weight
Liquid chemicals (HCl, etchants)	Tank truck or ISO tank	Hazardous material; specialized transport
Finished products (Aegis, Charlemagne)	LTL truck or parcel	Moderate size; moderate value

7.0 SUPPLY CHAIN RISK MANAGEMENT

7.1 Risk Register

Risk	Probability	Impact	Mitigation
MAX phase supplier failure (only 2-3 global suppliers)	MEDIUM	CRITICAL	In-house synthesis capability; strategic stockpile; fund supplier capacity
Basalt fiber supplier disruption (Russia-Ukraine conflict)	MEDIUM-HIGH	HIGH	Qualify US/EU suppliers; forward integration evaluation
BaCO ₃ supply disruption (China export restriction)	MEDIUM	MODERATE	Non-Chinese suppliers; stockpile; alternative formulations
Transportation disruption (port strike, canal blockage, rail strike)	LOW-MEDIUM	MODERATE	Multi-modal flexibility; regional warehousing
Natural disaster at manufacturing facility	LOW	HIGH	Geographic redundancy (multi-facility network)

Risk	Probability	Impact	Mitigation
Pandemic / workforce disruption	LOW-MEDIUM	MODERATE	Remote-capable functions; cross-training; safety stock
Cyber attack on supply chain systems	LOW-MEDIUM	MODERATE	Cybersecurity program (see DD-04); manual fallback procedures

7.2 Business Continuity in Supply Chain

CSM's supply chain business continuity plan includes: - **Supplier Failure Protocol:** 72-hour response: activate alternative supplier; expedite qualification if supplier previously unqualified; draw down safety stock; communicate impact to customers - **Logistics Disruption Protocol:** Re-route through alternative ports/carriers; expedited freight budget; customer communication - **Force Majeure:** Defined in all supplier contracts; mutual obligations during force majeure events

8.0 VERTICAL INTEGRATION ANALYSIS

8.1 Make vs. Buy Decision Matrix

Activity	Current Decision	Rationale	Re-evaluation Trigger
MAX phase synthesis	BUY (short-term); MAKE (evaluate for Phase 2-3)	Core IP dependency; supply base is thin	If commercial suppliers cannot meet quality/cost/volume requirements
Basalt fiber production	BUY	Capital-intensive; existing industry adequate	If geopolitical disruption removes major suppliers; if volume justifies dedicated line
CNT production	BUY	Capital-intensive; adequate supply base	If quality consistency becomes issue at volume
Polymer resin production	BUY (always)	Commodity; no strategic advantage in resin manufacturing	N/A
Geopolymer binder (alkali activator)	BUY	Commodity chemicals	N/A
Electronics manufacturing	BUY (always)	Contract manufacturing ecosystem mature	N/A
Final assembly and testing	MAKE (always)	Quality-critical; IP-sensitive	N/A

9.0 AGENT VOICES

Supply Chain Agent: “The MXene supply chain does not exist yet. There are no commercial-scale MAX phase producers. There are no large-scale MXene etchers. CSM is creating the MXene supply chain from scratch. That’s both terrifying (no existing infrastructure to leverage) and an opportunity (shape the supply chain to CSM’s specifications from the beginning). The MAX phase synthesis question — make or buy — is the single most important supply chain decision CSM faces. If we make it, we control our destiny but absorb capital and operational complexity. If we buy, we depend on suppliers who may not exist yet. Choose carefully.”

El Segundo (Resilience): “A supply chain that can’t survive the very event CSM protects against is a failed supply chain. The irony is not lost: CSM must build a grid-resilient supply chain to manufacture grid-resilient products. That means backup power at all facilities. EMP-hardened data centers. Geographically distributed warehouses. Supplier business continuity plans that account for GMD. The company that sells civilizational resilience must itself be resilient.”

The Accountant (Dual-Sourcing Economics): “Dual-sourcing costs more than single-sourcing. Two qualified suppliers means two audits, two quality programs, two sets of purchase orders — and less volume leverage with each. But dual-sourcing also means you survive when your single source has a fire, a flood, a bankruptcy, or a geopolitical crisis. The Accountant Heuristic says: quantify the cost of supply disruption. For CSM, a supply disruption during a Carrington warning would mean transformers fail that could have been saved. The cost is measured in billions and lives. Dual-sourcing is cheap insurance.”

Document prepared for investor and operational planning purposes. All supplier names are publicly available information. CSM does not have existing commercial relationships with named suppliers unless otherwise stated. Supply chain development is ongoing and subject to change.